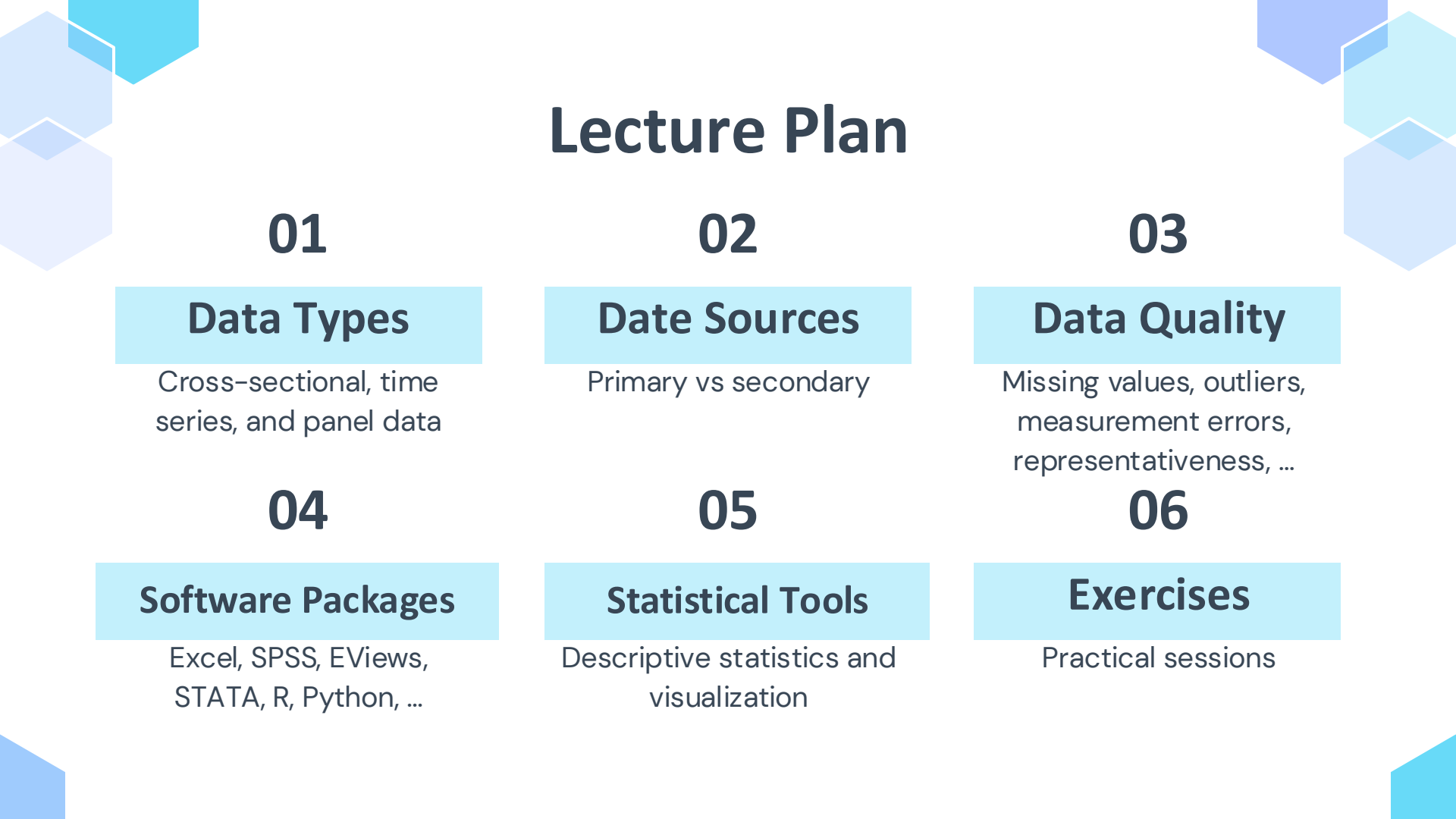


REM2 Research Methods

Lecture 3: Data Issues

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Lecture Plan

01

Data Types

Cross-sectional, time series, and panel data

02

Date Sources

Primary vs secondary

03

Data Quality

Missing values, outliers, measurement errors, representativeness, ...

04

Software Packages

Excel, SPSS, EViews, STATA, R, Python, ...

05

Statistical Tools

Descriptive statistics and visualization

06

Exercises

Practical sessions

Data Types - Univariate

Observation units (residents, houses, firms, countries, ...)

	House 1	House 2	House 3	House 4	House 5	House 6	House 7	House 8	House 9	House 10
2000	200	1500	444	360	1700	580	2200	400	300	1500

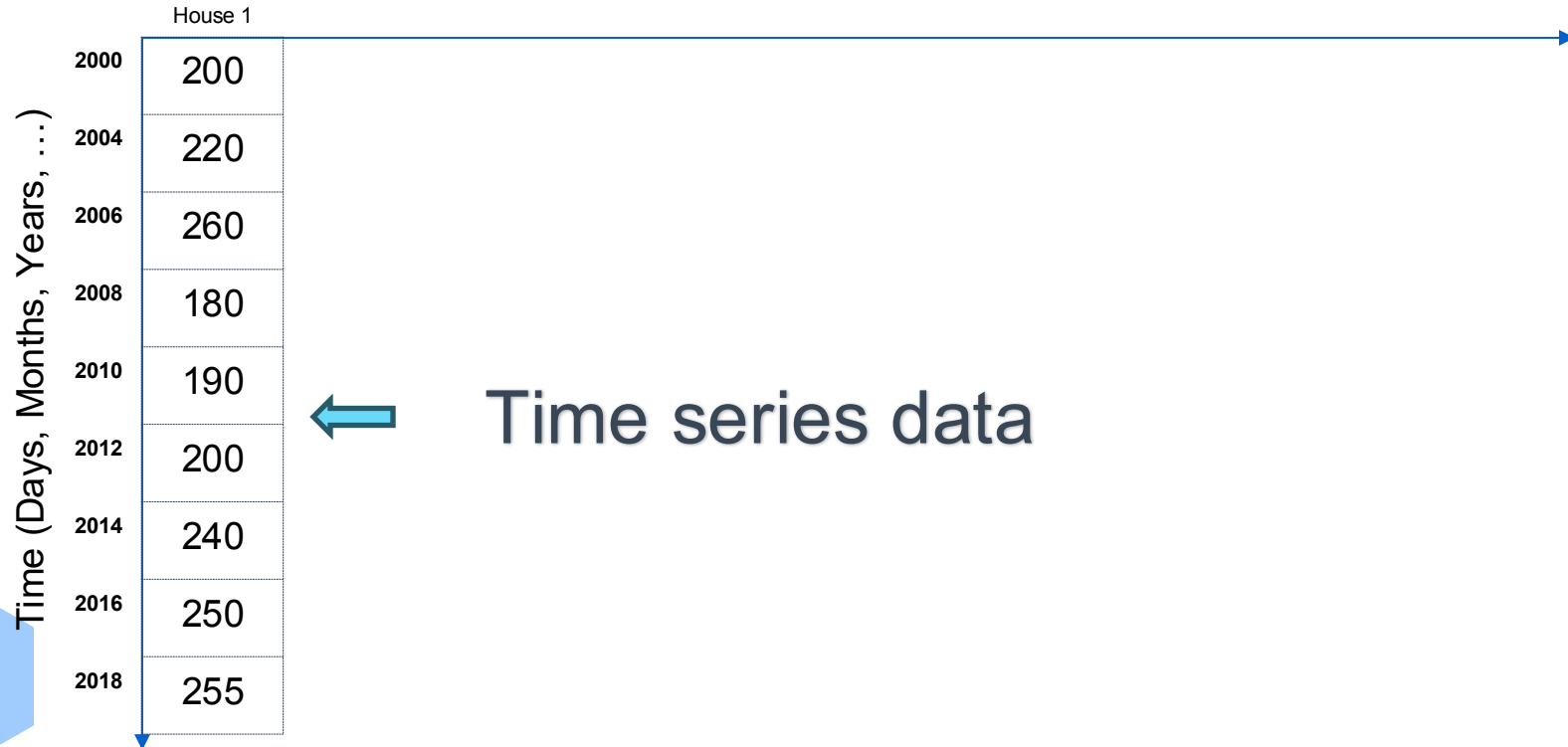


Cross-sectional data

Time (Days, Months, Years, ...)

Data Types - Univariate

Observation units (residents, houses, firms, countries, ...)



The diagram illustrates univariate time series data for a single observation unit, 'House 1'. A vertical table on the left lists data points from 2000 to 2018. A horizontal blue arrow points from the top of this table to the right, representing the cross-sectional dimension. A vertical blue arrow points downwards along the left side of the table, representing the time dimension. A teal arrow points from the text 'Time series data' to the table. The table has a light blue border and a light blue background.

	House 1
2000	200
2004	220
2006	260
2008	180
2010	190
2012	200
2014	240
2016	250
2018	255

Time series data

Data Types - Univariate

Observation units (residents, houses, firms, countries, ...)

Time (Days, Months, Years, ...)

	House 1	House 2	House 3	House 4	House 5	House 6	House 7	House 8	House 9	House 10
2000	200	1500	444	360	1700	580	2200	400	300	1500
2004	220	1800	440	400	1900	580	3000	400	360	1500
2006	260	1900	400	400	2200	580	2500	400	360	1000
2008	180	1000	400	400	2400	580	2500	300	360	1000
2010	190	1100	300	400	2800	580	2800	300	400	1400
2012	200	1500	300	500	3300	580	3200	300	500	1400

Panel data

Data Types – Multivariate

Variables (Rent, Returns, Prices, GDP, Interest Rates, ...)

	Variable 1	Variable 2	Variable 3	Variable 4	Variable 5	Variable 6	Variable 7	Variable 8	Variable 9	Variable10
Period 1										
Period 2										
Period 3										
Period 4										

Time (Days, Months, Years, ...)

Panel data

Data Types – Multivariate

Variables (Rent, Returns, Prices, GDP, Interest Rates, ...)

	Variable 1	Variable 2	Variable 3	Variable 4	Variable 5	Variable 6	Variable 7	Variable 8	Variable 9	Variable10
Period 1										
Period 2										
Period 3										
Period 4										

Time (Days, Months, Years, ...)

Panel data

Data Types – Multivariate

Variables (Rent, Returns, Prices, GDP, Interest Rates, ...)

	Variable 1	Variable 2	Variable 3	Variable 4	Variable 5	Variable 6	Variable 7	Variable 8	Variable 9	Variable 10
Period 1										
Period 2										
Period 3										
Period 4										

Time (Days, Months, Years, ...)

Panel data

Data Types – Multivariate

Variables (Rent, Returns, Prices, GDP, Interest Rates, ...)

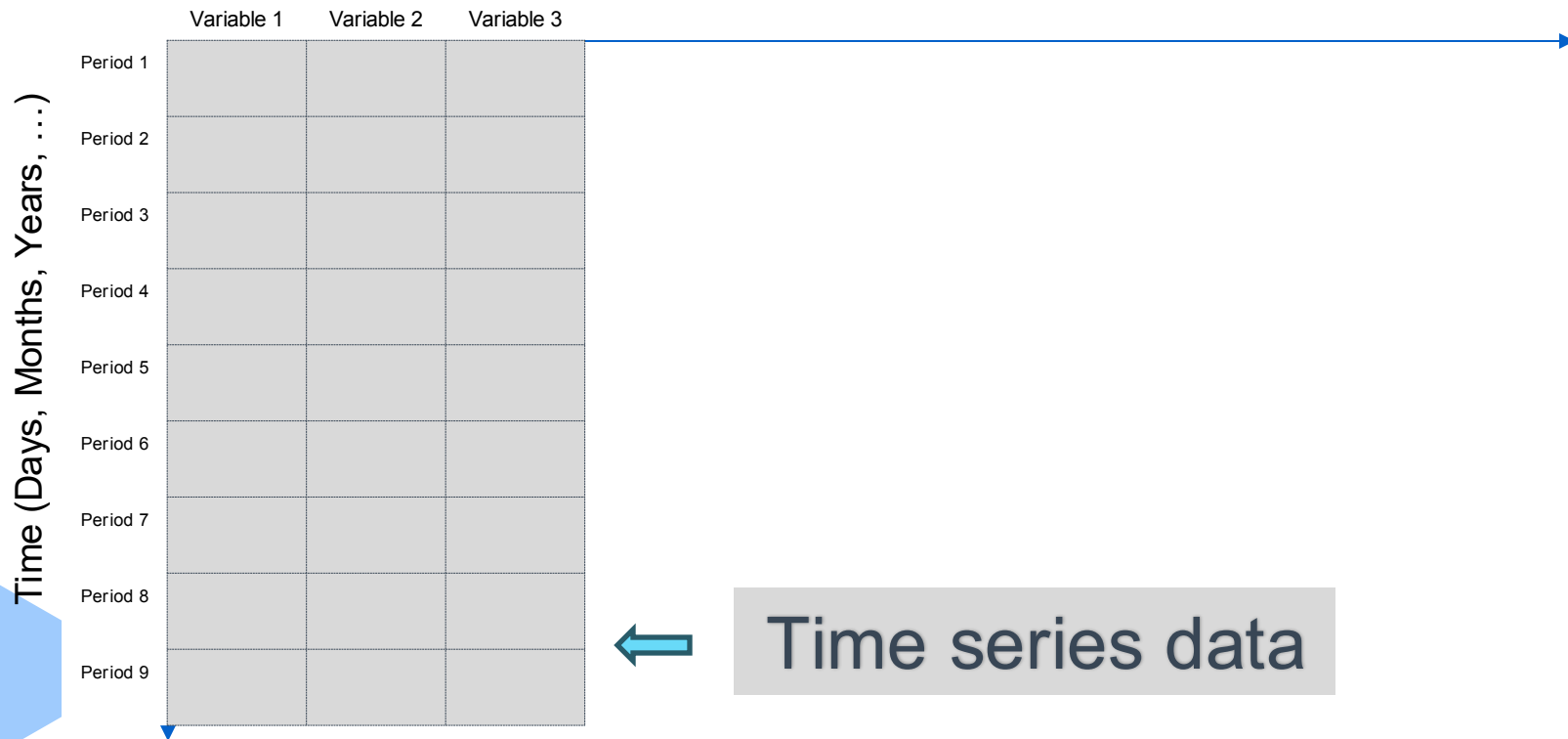
	Variable 1	Variable 2	Variable 3	Variable 4	Variable 5	Variable 6	Variable 7	Variable 8	Variable 9	Variable10
Period 1										
Period 2										
Period 3										
Period 4										

Time (Days, Months, Years, ...)

Panel data

Data Types – Multivariate

Variables (Rent, Returns, Prices, GDP, Interest Rates, ...)



What's the difference?

Data types



Cross-sectional

Observations on multiple objects at one point in time



Time series

Observations on one object over multiple points in time



Panel data

Observations on multiple objects over multiple points in time

Data analysis methods



Cross-sectional

OLS, GMM, DID, RDD, ...



Time series

AR, VAR, ECM, GARCH, ...



Panel data

Pooled, FE or RE panel regression, Dynamic panel regression, ...

Data Sources

Primary Data

- Collected firsthand by the researcher for a specific research purpose
- Collected through surveys, experiments, interviews, or direct observations
- More control and specificity but can be expensive and time-consuming

Secondary Data

- Has already been collected by someone else and is available for use by researchers
- Existing databases, official reports, published studies, and archival records
- More accessible and cost-effective but may not be tailored to the exact needs of the research

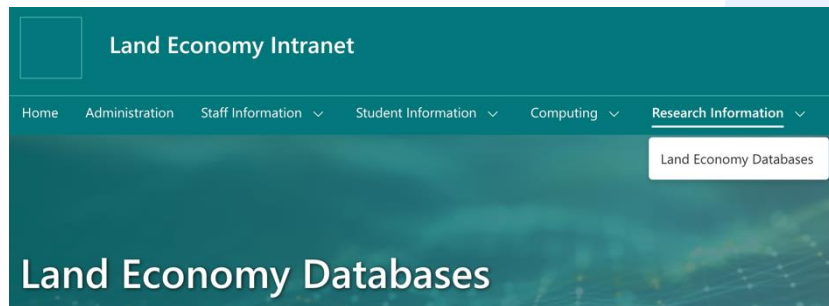
Quality concerns when using secondary data

Concerns	Consequences	Mitigations
Missing values	Biased results if not properly handled. The extent and pattern of missing data (random or systematic) also affect the validity of the analysis.	Imputation, deletion of missing cases, or using methods like maximum likelihood estimation
Outliers	Distortion of patterns, especially in small datasets	Removal (be careful! Some are legitimate) Sensitivity analysis to determine influence.
Measurement errors	Caused by inconsistencies in data collection methods, coding errors, or changes in definitions over time. Inaccurate results.	Cross-validate with other sources. Apply correction techniques if the error structure is known.
Representativeness	Over- or under-sampling of certain groups or lack sufficient geographic or demographic coverage.	Weightings. Carefully interpreting and generalizing findings.

Land Economy Databases (Moodle and Intranet)

Bloomberg (Public Equity – Broad)

Source	Bloomberg
Type	Public Equity
Category	Broad
Description	Platform containing a plethora of financial statistics.
Website	http://www.bloomberg.com/
Access	Access to Bloomberg is via a computer terminal in Judge Business School. Instructions for use are on the terminal. See also: https://www.jbs.cam.ac.uk/faculty-research/information-library-services/databases/



CoStar (Private Equity – Commercial - US & UK)

Source	CoStar
Type	Private Equity
Category	Commercial - US & UK
Description	CoStar gathers data from eight critical sources and delivers it over a single, easy-to-use platform.
Website	http://www.costar.co.uk/
Access	Land Economy now has global access to the US and other countries (i.e. Canada and the UK) data. Use your CRSID@cam.ac.uk email address only to register via the link below (it will not accept a personal email address. The form asks for the name of a tutor, so put Zilong Wang in this section. https://costar.na1.echosign.com/public/esignWidget?wid=CBFCIBAA3AAABLb1qZhCkstOXdj1WiWzdCNbb1pfEpumHUOX9ObCG36IC5xouS5nrXG5DJPvOVJcrCpTIU*

BHPS (Other – UK)

Source	BHPS
Type	Other
Category	UK
Description	The British Household Panel Survey began in 1991 and is a multi-purpose study whose unique value resides in the fact that it follows the same representative sample of individuals over a period of years.
Website	https://www.iser.essex.ac.uk/bhps See also: https://www.iser.essex.ac.uk/bhps/acquiring-the-data
Access	Refer to website for further details.

Statistical software packages

Available for free for CAM students	Pros and Cons	Best at ...
Excel 	Great for basic analysis and data manipulation, but limited for advanced quantitative analysis. Up to 16 independent variables in regression analysis.	Checking outliers and missing values Preliminary analysis (charts and tables) Descriptive statistics Visualisation of results
SPSS 	User-friendly and ideal for social sciences, but limited in flexibility and advanced methods. https://cambridge.store.academia.co.uk/	Standard regression analysis Not very good at time series analysis GUI, not much coding options
STATA 	Excellent for econometrics and handling large datasets, but costly and not as flexible as R for customization. https://www.it.econ.cam.ac.uk/stata	Standard and advanced regression analysis and time series analysis Write your own codes Run models/analyses in batches
R 	Highly flexible, powerful, and open-source, but requires programming knowledge and has a steeper learning curve. https://posit.co/download/rstudio-desktop/	Write your own codes Run models/analyses in batches Visualisation Advanced and new methods (ML, LLM, ...)

Statistical Tools



Descriptive

Mean/Median/Mode,
Range/Variance/SD
Max & Min

Correlation matrix / Scatter plots
Time series charts

... ..

This lecture



Inferential

Multiple linear regression
Difference-in-Differences
Regression discontinuity
design
Panel regression

... ..

Lectures 3, 5 and 6



Predictive

Exponential smoothing
Moving average
Autoregressive regression
Neural network
Machine learning

... ..

Exercise 1: Inflation and REITs returns

Instructions: Work in groups of two to complete the following tasks

1. Identify the data sources of the key variables – Anxious Index, realised and expected CPI inflation rates, and REITs returns in Connolly and Stivers (2024).
2. Download the Anxious Index (AI) and Median CPI Inflation Rate (CPI) from the Survey of Professional Forecasters (SPF) website.
3. Open the downloaded Excel files. Observe the data. Note any missing values or outliers. Find the variable definitions in each file (Hint: Documentations).
4. Determine the type of the data - are these cross-sectional, time series, or panel data?
5. Identify the statistical methods used in Connolly and Stivers (2024).
6. Use a software package of your choice, generate descriptive statistics for the downloaded variables.
7. Use a software package of your choice, visualise the trends of Anxious Index and Expected Inflation Rate.
8. If you want to replicate this analysis by using data from the UK, can you find the corresponding data sources from the CRERC Databook? Discuss.
9. To estimate the model on page 929, which of the four software packages introduced in this lecture could be used? Discuss.

Exercise 2: Shrinking Homes in London

Instructions: Work in groups of two to complete the following tasks

1. Identify the data sources of the key variables – property size from EPC and property transaction records in Hubbard et al, (2024).
2. Download the EPC data from the Department of Levelling up, Housing & Community's website.
3. Download the property transaction data from the HM Land Registry Open Data website.
4. Open the downloaded files in Excel. Observe the data. Note any missing values or outliers. Find the variable definitions in each file (Hint: Documentations).
5. Determine the type of the data - are these cross-sectional, time series, or panel data?
6. Use a software package of your choice, generate descriptive statistics of the variables in the two datasets.
7. Use a software package of your choice, visualise the distribution of house prices (i.e., histograms or pie charts) in London.
8. If you want to replicate this study by using data from Cambridge, can you find the data to support your analysis? Discuss.

Further Questions?

